qPCRtools: a cross-platform C++ desktop application for qPCR data analysis with validated equivalence to its R implementation

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Abstract

Quantitative real-time PCR (qPCR) is a standard technique for measuring gene expression, yet turning instrument-exported C_q values into publication-ready results requires a chain of non-trivial steps: amplification-efficiency estimation, choice of quantification method, statistical comparison of groups, and figure preparation. The established tools for this chain are predominantly programming-based—including our own R package *qPCRtools*—and therefore remain out of reach for part of the laboratory community, notably students and wet-lab researchers without programming experience, whereas commercial alternatives are costly, closed-source, and platform-restricted. Here we present *qPCRtools*, a cross-platform desktop application written in C++17 with Qt6 that exposes the analytical core of the R package through a point-and-click interface and ships as a single executable requiring no R, Python, or other runtime environment. The application implements standard-curve construction and amplification-efficiency estimation, the ΔC_t, $\Delta\Delta C_t$ and efficiency-based (RqPCR) relative quantification methods, group comparison by two-sample t-test, Wilcoxon rank-sum test, or one-way ANOVA with Tukey's HSD letter grouping, and one-click export of both the summary table and

the per-replicate raw data. We validated the C++ implementation against the R package on identical inputs: across every shared quantification method the compared quantities agree to floating-point precision, and the validation is reproducible from the open scripts shipped with the software. *qPCRtools* is freely available under the MIT license for Windows and macOS, and its source code compiles on Linux.

1 Introduction

Quantitative real-time PCR (qPCR) is among the most widely used techniques in molecular biology for quantifying nucleic acids and measuring gene expression [5]. A sound qPCR analysis involves several interdependent steps: estimating primer amplification efficiency, choosing an appropriate quantification strategy, performing statistical testing, and producing publication-quality figures [4]. When the amplification efficiencies of target and reference genes are comparable, the $2^{-\Delta\Delta C_T}$ method [8] is convenient; when they differ, an efficiency-corrected or standard-curve-based approach is required instead [11]. These analytical choices, combined with the need for rigorous statistics and clear visualization, make qPCR data analysis demanding for many laboratory researchers.

A range of tools address qPCR analysis [10], yet each covers a different part of the workflow (Table 1). Scripting-based implementations—our own *qPCRtools* [7] and *pcr* [1]—implement the calculation chain transparently, but require programming skill and a configured software environment. Software distributed with real-time instruments (for example QuantStudio Design & Analysis and CFX Maestro) offers a graphical workflow but is tied to that instrument platform. Free stand-alone programs cover individual steps: REST [12] computes efficiency-based relative expression for two-group comparisons with a randomisation test, LinRegPCR [13, 15] estimates baseline and amplification efficiency from raw fluorescence, and qBase [6] implements relative quantification together with the geNorm stability measure [17] for selecting reference genes. Free web applications integrate several of these steps—SATQPCR [14], DAG Expression [3] and Auto-qPCR [9]—but they require uploading the data to a remote server (or deploying the web stack locally), which constrains offline use and the handling of unpublished data. The gap we address is therefore not a missing algorithm but a missing delivery mode: a tool that combines *validated numerical equivalence* to a published implementation with a *native, cross-platform, zero-dependency* graphical workflow.

Table 1: Delivery mode and functionality of representative qPCR analysis tools. • provided, ○ not provided, ○/• partial or version-dependent (statistics, for example, were added in the later qBase+ release, and vendor software differs between instruments and versions). Feature sets were compiled from the cited papers, the vendors' documentation and the survey of Pabinger *et al.* [10]. [a] *qPCRtools* [7], *pcr* [1]. [b] e.g. QuantStudio Design & Analysis, CFX Maestro. [c] SATQPCR [14], DAG Expression [3], AutoqPCR [9]. REST [12], LinRegPCR [13, 15], qBase [6], geNorm [17].

Tool	Free	No coding	Offline	Efficiency	Statistics	Ref. genes
R packages[a]	•	○	•	•	•	•
This application	•	•	•	•	•	•
REST	•	•	•	•	•	○
LinRegPCR	•	•	•	•	○	○
qBase	•	•	•	•	○/•	•
Instrument software[b]	•	•	•	•	○/•	○
Web apps[c]	•	•	○	•	•	•

Statement of need

The programming requirement is not a marginal obstacle. In our own teaching and collaborative work, the users who most need a defensible qPCR analysis are frequently the ones least able to install an R environment, resolve package dependencies, and write the few lines of code that the analysis requires: newly enrolled students, visiting researchers, and wet-lab staff whose projects run for a single season. They currently fall back either on manual spreadsheet calculations—which are difficult to audit and easy to get wrong—or on whichever commercial instrument software happens to be installed. Our R package [7] already provides the algorithms, and its adoption since 2022 (more than 30 citations) shows that the analytical choices it implements are accepted in published work, but that adoption is bounded by the R environment itself.

We therefore reimplemented the analytical core as a native desktop application. The design goal is deliberately narrow: keep the validated behaviour of the R implementation, and remove every prerequisite except a double-click.

We introduce *qPCRtools*, a C++/Qt6 desktop application that reimplements the analytical core of our R package behind a point-and-click interface distributed as a single executable. This Application Note makes three contributions: (i) a C++/Qt architecture in which a tested core algorithm library is decoupled from a web-based user interface; (ii) a validation demonstrat-

ing numerical equivalence to the R package across all shared quantification methods, shipped as a reproducible validation package that a reader can re-run; and (iii) statistical and functional extensions beyond the original R version, including non-parametric tests, effect sizes, outlier removal, and automatic reference-gene selection. Preparing the port also revealed defects in result aggregation that are shared with the R package—the input of the reference-gene stability measure, replicate handling in the efficiency method, and the $\Delta\Delta C_t$ calibrator—as well as one specific to the C++ implementation (multiple-comparison p-values); all were corrected and are covered by regression tests.

2 Implementation

2.1 Architecture

qPCRtools is written in C++17 and built on Qt6 [16], which provides cross-platform abstractions and the native application shell. The user interface is rendered with web technologies—Bootstrap 5.3 for layout, with Apache ECharts [2] bundled for the figure view—hosted inside a `QWebEngineView`, and exchanges data with the native backend through Qt WebChannel via a registered bridge object. This separation lets the presentation layer iterate rapidly as HTML, CSS, and JavaScript while all numerical computation stays in compiled native code. The analytical core is isolated as a standalone static library, `qpcr_core`, which depends only on Qt Core and Eigen and is independent of the GUI; it is exercised directly by the unit-test suite and is also linked by the command-line equivalence driver described in Results. Statistical distribution functions use the GNU Scientific Library (GSL) for accuracy. The resulting interface is shown in Figure 1.

2.2 Quantification algorithms

The core reimplements the analytical methods of the R package. From a dilution series, amplification efficiency is obtained by ordinary least squares of C_q on $\log_d$(concentration), i.e. on the number of d-fold dilution steps (d = dilution factor, 4 in the bundled example), yielding slope s, intercept, R^2, and a regression p-value; efficiency is

$$E = d^{-1/s} - 1, \tag{1}$$

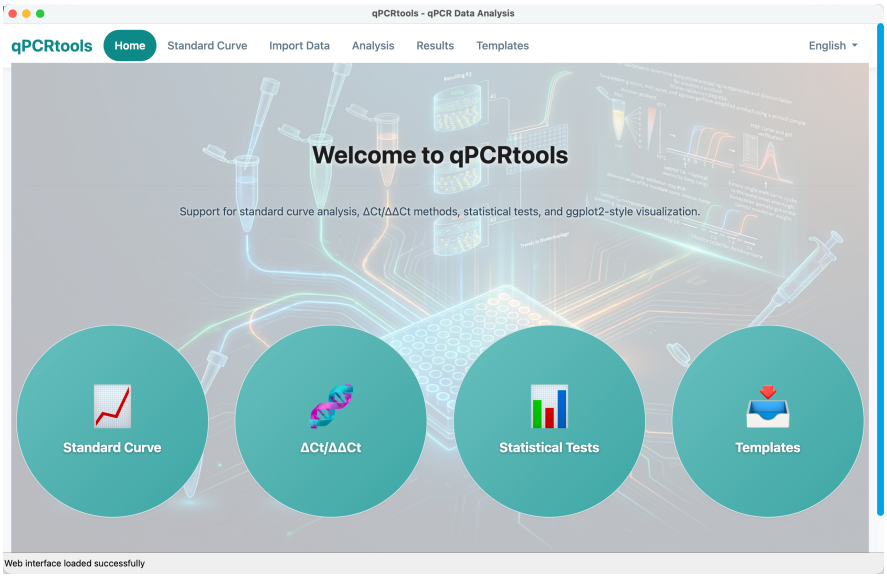


Figure 1: The *qPCRtools* desktop interface (version 1.1.0; screen captured from the running application). A single cross-platform window hosts the web-based UI: the home screen gives access to the four analysis areas (standard curve, $\Delta C_t/\Delta\Delta C_t$, statistical tests, and templates for data entry). The interface is bilingual (Chinese/English); the English language is shown here.

with s expressed per $\log_d$ unit (equivalently, per dilution step). Relative expression is computed by the ΔC_t and $\Delta\Delta C_t$ methods [8],

$$\mathrm{RE} = 2^{-\left[(C_{q,\mathrm{target}} - C_{q,\mathrm{ref}})_{\mathrm{treated}} - (C_{q,\mathrm{target}} - C_{q,\mathrm{ref}})_{\mathrm{control}}\right]}, \tag{2}$$

and by an efficiency-based (RqPCR) method [14] in which per-sample quantities are $E^{\,C_{q,\min} - C_q}$ and are normalized by the geometric mean of one or more reference genes [11].

2.3 Statistical analysis

For each target gene the application compares groups statistically. The core library implements two-sample t-tests with and without the Welch variance correction, the paired t-test, the Wilcoxon rank-sum and signed-rank tests, and one-way ANOVA followed by Tukey's HSD post-hoc letter grouping, and it reports Cohen's d effect sizes alongside the t-test results; it also supports IQR-based outlier removal and automatic reference-gene selection by the

geNorm algorithm [17]. These options extend the original R package, which provided t-test and ANOVA/Tukey testing only. The graphical interface currently exposes the Welch two-sample t-test, the Wilcoxon rank-sum test and one-way ANOVA with Tukey's HSD; the paired t-test and the Wilcoxon signed-rank test are exercised by the test suite but are not yet offered as interface options.

2.4 Software engineering and scope

All results are presented as tables that can be exported as CSV or Excel for further plotting, and groups that differ significantly are labelled with Tukey letters directly in the result table. Version 1.1.0 does not render figures: the ECharts-based figure view is present in the code base but disabled, and the figures of this paper were therefore produced from the application's numerical output by the plotting script shipped in the repository (`validation/plot_examples.py`), not by the graphical interface. *qPCR-tools* is managed with CMake and ships with a unit-test suite; the interface is bilingual (Chinese/English) and reads CSV input as well as Excel (`.xlsx`) workbooks when built against the optional OpenXLSX dependency. The application is distributed as a single self-contained executable for Windows and macOS, requiring no R, Python, or runtime installation, and the source compiles on Linux. It is released under the MIT license. Two R functions are not yet ported—the standard-curve slope–intercept quantification (`CalExpCurve`) and the reverse-transcription reagent volume calculation (`CalRTable`)—and are planned for a future release.

3 Results

3.1 Typical analysis workflow

To illustrate a typical analysis, we analyzed the bundled example dataset (`examples/cq.csv` and `examples/design.csv`), comprising the target gene *fos-glo-myc* normalized against the reference gene *Beta Actin* across three treatment groups (0, 0.5, 1) with three biological replicates each. Using the $\Delta\Delta C_t$ method with a Welch two-sample t-test against the control group (0), relative expression rose monotonically to 1.92 at dose 0.5 and 2.74 at dose 1, both significantly different from the control ($p = 0.006$ and $p = 0.022$; Figure 2). As defined by Livak and Schmittgen, the calibrator is the control group, so the control group is ≈ 1 (here 1.01 ± 0.21). The group means, standard deviations and p-values quoted here are repro-

duced without any dependency on the application or on the R package by `validation/check_example_stats.R`. Applied to a four-fold dilution series of four genes, the standard-curve module recovered slopes between -2.07 and -2.24 C_q per four-fold dilution step ($\log_4$ scale) with $R^2 > 0.99$ for three of the four primers (efficiencies 0.85–0.95), illustrating amplification-efficiency estimation that, per MIQE guidance, determines which quantification method is appropriate. The complete workflow—import, configuration, analysis, and export—is performed through the graphical interface without writing code.

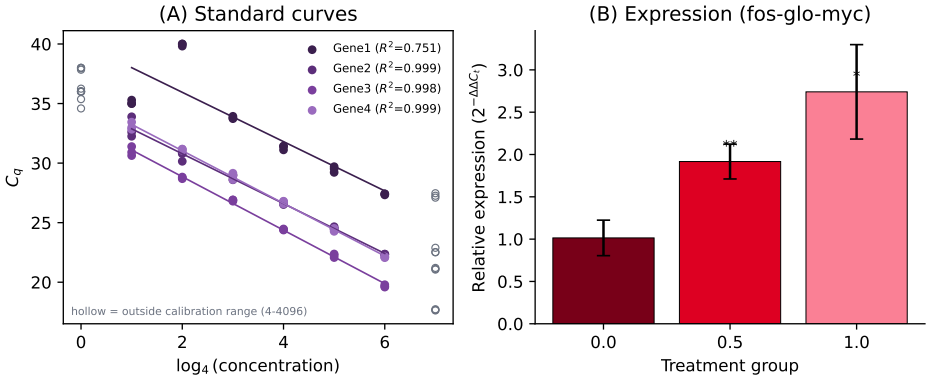


Figure 2: Analysis output on real data. (A) Standard curves for four genes over a four-fold dilution series, with fitted regressions and R^2; the abscissa is $\log_4$ concentration (dilution steps), matching the regression, and hollow symbols mark wells outside the calibrated range (4–4096) that are excluded from the fit. (B) Relative expression of *fos-glo-myc* across treatment groups ($\Delta\Delta C_t$ method, calibrated to the control group), with significance labels. Both panels were drawn from the numerical output of the application (the standard-curve parameters and the per-group means exported by the results table) by the plotting script shipped in the repository; version 1.1.0 exports result tables rather than rendering figures.

3.2 Equivalence to the R package

We validated the C++ implementation against the original R package [7] by running both on identical inputs for every shared method—$\Delta\Delta C_t$ (`CalExp2ddCt`), the standard curve (`CalCurve`), and the efficiency-based RqPCR method (`CalExpRqPCR`)—and comparing every resulting quantity

pairwise (Table 2, Figure 3). Inputs were drawn from the repository's example data and from a dilution series used in the original R package; both implementations read the same on-disk files, so any difference reflects the implementations alone.

Table 2: Equivalence of C++ and R results across all shared methods on identical inputs. Relative difference is $|R - \text{C++}|/|R|$.

Method	Compared quantities	Pairs	Max. rel. diff.
$\Delta\Delta C_t$	expression, SD	6	3.8×10^{-15}
Standard curve	slope, intercept, R^2, efficiency	16	2.5×10^{-3}
RqPCR	expression, SE	16	2.1×10^{-15}

For the two expression methods, $\Delta\Delta C_t$ and RqPCR, every compared quantity—expression values, standard deviations, and standard errors—matched between implementations to within floating-point precision (maximum relative difference $< 4 \times 10^{-15}$). For the standard curve, slope, intercept, R^2, and efficiency agreed to within 2.5×10^{-3} relative difference; the residual is attributable to the R package rounding its tabulated output to two to four decimal places, since the underlying C_q inputs are identical. Plotting every C++ result against its R counterpart, all points fall on the line $y = x$ (Figure 3). These results confirm that the C++ reimplementation preserves the analytical behavior of the R package across all shared methods. The validation is fully reproducible from the open scripts in `validation/` (see Availability).

4 Availability

Version 1.1.0 of *qPCRtools* is described here. The source code and pre-built binaries for Windows and macOS are available at `https://github.com/lixiang117423/qPCRtools_cpp` under the MIT license; no registration is required to download or run the software. The repository includes documentation, a unit-test suite, example datasets, and a fully reproducible equivalence-validation package (`validation/`) comprising the R script, a C++ command-line driver linked against `qpcr_core`, and a comparison script that regenerates all validation outputs and figures. The project builds with CMake ($\geq$ 3.20), requires a C++17 compiler and Qt 6 ($\geq$ 6.5) with the WebEngine and WebChannel modules, and optionally links Eigen3, GSL, and OpenXLSX.

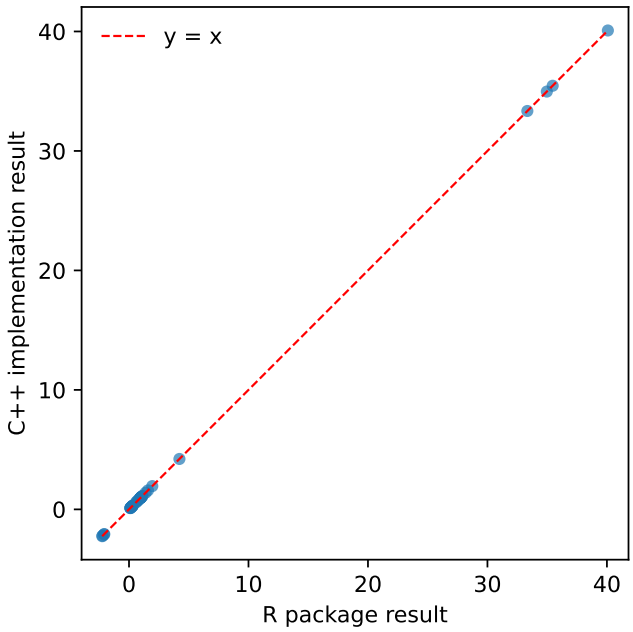


Figure 3: Numerical equivalence of the C++ implementation and the R package. Each C++ result is plotted against the corresponding R result across all shared methods ($\Delta\Delta C_t$, standard curve, RqPCR); all points fall on the $y = x$ line.

Authors' contributions

XL and HH designed this study and performed the analysis. XL wrote the manuscript. Both authors read and approved the final manuscript.

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Conflict of interest

The authors declare no competing interests.

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